

Phase 3 – Project Design

Introduction

This phase describes the overall design of the Personalized Networking Assistant, including system architecture, modules, data flow, and design decisions.

High-Level Design

The application follows a modular architecture with a Streamlit frontend, FastAPI backend, AI services (DistilBERT and GPT-2), Wikipedia API integration, and local JSON storage for history and feedback.

Low-Level Design

Frontend handles user interactions. FastAPI routes requests to services. Theme Extractor analyzes events, Starter Generator creates conversation starters, Fact Checker validates information, and History Logger stores interactions.

System Architecture

User → Streamlit UI → FastAPI Backend → AI Models (DistilBERT & GPT-2) → Wikipedia API → JSON Data Store → Response to User.

Data Flow

1. User enters event details.
2. Backend extracts themes.
3. GPT-2 generates conversation starters.
4. User may verify facts.
5. Feedback and history are stored.

Design Objectives

Maintain modularity, scalability, reliability, fast response time, and easy maintenance while keeping AI services reusable.

Module	Purpose
Frontend (Streamlit)	Collect user input and display AI-generated results
Backend (FastAPI)	Process requests and orchestrate services
Theme Extractor	Identify event themes using DistilBERT
Starter Generator	Generate personalized conversation starters using GPT-2
Fact Checker	Verify claims using Wikipedia API
History & Feedback	Store interactions and user ratings